BanglishCheck

Explainable AI-Generated Text Detection for Banglish

Technical Report — Research Den Fall 2026 Technical Project

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Abstract

Automatic AI-text detectors are mostly built and tested on English, but a large share of everyday online writing in South Asia is Banglish: Bangla written in the Latin alphabet, often mixed with English in the same sentence, with no fixed spelling for the same word. BanglishCheck builds a detector for Banglish specifically and treats explanation as a first-class requirement: every prediction comes with the words and writing-style signals that drove it, so a user can judge the score instead of trusting it blindly. A classical TF-IDF plus logistic-regression model, augmented with 14 hand-built stylometric features, reaches 96.5% accuracy and 0.996 AUROC on a held-out test set. A deletion test supports the faithfulness of its explanations: removing the five words ranked highest by the explainer drops confidence by 0.45 on average, compared with 0.15 for five random words. A leave-one-generator-out test also exposes the main limitation: performance falls on writing from an unseen generator, so the detector must be treated as decision support rather than proof of authorship.

1. Introduction

Bangla speakers increasingly write online in Banglish — Bangla transliterated into Latin script, frequently code-switched with English within a single sentence or even a single word. This is common on e-commerce review sites, social media and messaging, and it has no standard spelling: the same word ("valo", "bhalo", "balo" — "good") can appear several different ways in the same dataset. Since large language models can now write fluent Banglish, there is a growing need to tell apart human- and AI-written Banglish text — for moderation, for academic-integrity checks, and for platforms that want to label AI-generated content.

Existing AI-text detectors are built and validated almost exclusively on English, and they generally give a bare score with no explanation. Applying an English detector to Banglish risks silent failure: the detector may be keying on surface patterns (spelling regularity, sentence length) that mean something different in Banglish than in English. A bare score is also risky on its own: a false "AI-generated" label can be used to wrongly accuse a real person, so a detector that cannot show its evidence is harder to use responsibly.

This project asks two questions. First, can a Banglish-specific detector, built with Banglish linguistic and stylometric features, reliably separate human- from AI-written Banglish text? Second, can that detector explain itself — at the word level and at the level of writing-style signals — in a way that is demonstrably faithful to what the model actually used, not just plausible-looking?

2. Related Work

AI-text detection is usually framed either as supervised classification, using labeled human and AI examples, or as a zero-shot statistical task based on measures such as perplexity or token probability. This project takes the supervised

route with a classical TF-IDF and logistic-regression pipeline. The accompanying notebook also provides an XLM-RoBERTa experiment for future comparison once GPU results are available.

XLM-RoBERTa (Conneau et al., 2020) is a multilingual transformer pretrained on 100 languages, including Bengali, and is therefore a useful future baseline despite not being pretrained specifically on romanized, code-mixed Banglish. For interpretability, the deployed linear model uses exact SHAP values (Lundberg and Lee, 2017). The notebook also includes Integrated Gradients (Sundararajan et al., 2017), via Captum, for the planned transformer comparison. The interactive demo is implemented with Gradio (Abid et al., 2019) for deployment on Hugging Face Spaces.

The human side of the dataset comes from BanglishRev (Shamael et al., 2024), a large-scale dataset of Bangla-English code-mixed e-commerce product reviews — see Section 3 and the References.

3. Data

The dataset combines human-written Banglish product reviews with AI-generated Banglish e-commerce reviews from three generators:

Source	Texts	Notes
Human	1,638	Banglish product reviews from BanglishRev, posted before 30 Nov 2022 (ChatGPT's release)
ChatGPT	783	Generated Banglish e-commerce reviews
Claude	667	Generated Banglish e-commerce reviews
Gemini	188	Generated Banglish e-commerce reviews

- Reviews written in Bangla script, under 5 words, or over 60 words are removed.
- The human texts are sampled so their word-count distribution matches the AI texts', so the model cannot separate the two classes by length alone (see Figure 1).
- The data is split 70% train / 15% validation / 15% test, keeping each generator's share constant across splits.
- Human texts are dated before ChatGPT's public release so they cannot themselves be AI-generated.
- The raw data is not included in the project repository.

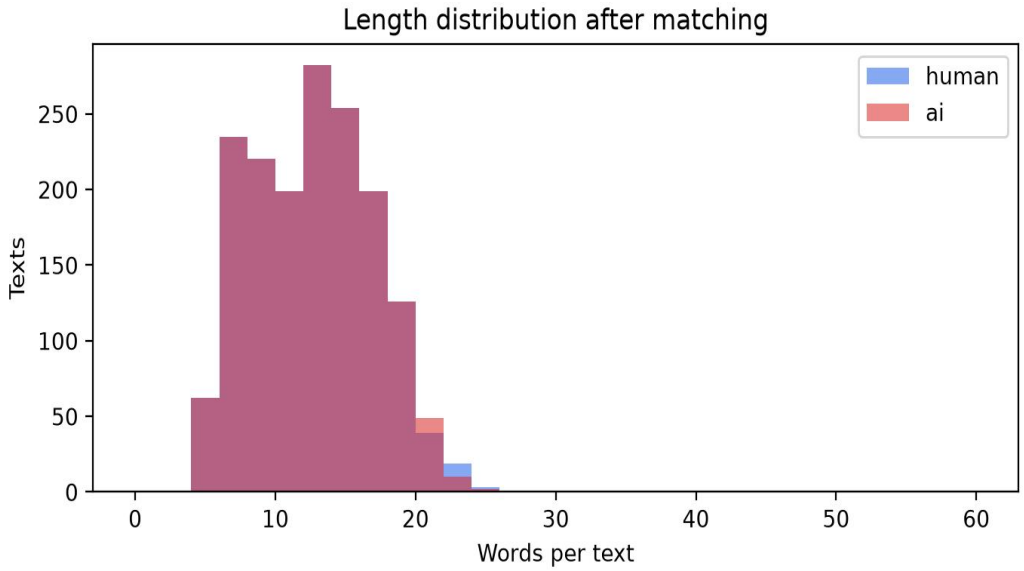


Figure 1. Word-count distribution of human vs. AI texts after length matching.

4. Methods

4.1 Feature extraction

Besides word (1–2 gram) and character (2–5 gram) TF-IDF features, 14 hand-built stylometric features target properties specific to Banglish and to AI-generated text generally: Banglish/English word ratio, code-switch rate (how often consecutive words alternate language), an informal-spelling-variant ratio (built from a hand-curated list of common Banglish spelling variants, e.g. valo/bhalo/vlo), punctuation and repeated-punctuation rate, emoji rate, word elongation ("sooo"), and sentence length, among others.

4.2 Models

The evaluated detector and the planned transformer comparison are:

- Classical (deployed): word n-grams + character n-grams + the 14 stylometric features, feeding a logistic regression classifier (scikit-learn FeatureUnion + LogisticRegression, class-balanced).
- Transformer (future comparison): XLM-RoBERTa base, configured for fine-tuning on the same training split in a Colab T4 GPU runtime. Its results are not reported because that run has not yet been completed.

The classical model was chosen for deployment because it performs strongly on the held-out data, runs on free CPU hardware, and, because it is linear, supports exact feature attributions. These practical advantages make it suitable for a reproducible student project even before a transformer comparison is available.

4.3 Explanations

For the deployed linear model, SHAP values are computed in closed form: for a linear model with feature independence, the SHAP value of feature j is coef_j x (x_j - mean_j). Word- and character-n-gram contributions are mapped back onto the words of the original text so each word receives a signed score (red pushes toward AI; blue pushes toward human). Stylometric contributions are reported by name in a bar chart. The notebook contains an Integrated Gradients workflow for explaining XLM-R predictions when that experiment is run.

4.4 System and deployment

The Colab notebook covers data cleaning and length matching, train/validation/test splitting, classical model training, robustness and faithfulness tests, figure generation, and demo artifacts. It also contains an optional XLM-RoBERTa training and Integrated Gradients workflow. The demo is a Gradio app that loads banglishcheck.py and explainer.joblib on a free Hugging Face Space; a standalone FastAPI version is included for self-hosting.

Tech stack:

Part	Tools
Data and features	Python, pandas, regular expressions, custom Banglish word lists
Classical models	scikit-learn (TF-IDF, FeatureUnion, LogisticRegression)
Explanations	Linear SHAP (exact), Captum Integrated Gradients
Transformer	Hugging Face Transformers, XLM-RoBERTa base, Google Colab T4 GPU
Demo	Gradio 6 on Hugging Face Spaces, and a standalone FastAPI version

5. Results

On a held-out test set of 492 texts (246 human and 246 AI), the full classical model substantially outperforms length-only and style-only baselines. Character n-grams alone obtain the highest accuracy and AUROC, while the full model obtains the highest AI-catch rate at the selected false-flag threshold and provides richer explanations:

Model	Accuracy	AUROC	AI caught at 5% false flags
Length only	61.0%	0.679	16.3%
Style features only	87.4%	0.942	72.8%
Word n-grams + LR	96.1%	0.995	97.2%
Char n-grams + LR	97.4%	0.998	98.8%
Full classical (deployed)	96.5%	0.996	99.6%
XLM-R (planned)	Not run	Not run	Not run

The XLM-RoBERTa row is marked as not run. Completing that experiment on the same split is future work; no transformer comparison is inferred from the classical results.

"AI caught" is measured at a decision threshold chosen so that only 5% of human validation texts are flagged — i.e., it reports how much AI text is caught while keeping false accusations of real people rare. The character-n-gram model alone is nearly as strong as the full model, suggesting most of the signal is in spelling/character-level patterns rather than word choice or style features in isolation — though the full model still gives the best AI-catch rate and, importantly, the richer explanations (Section 6).

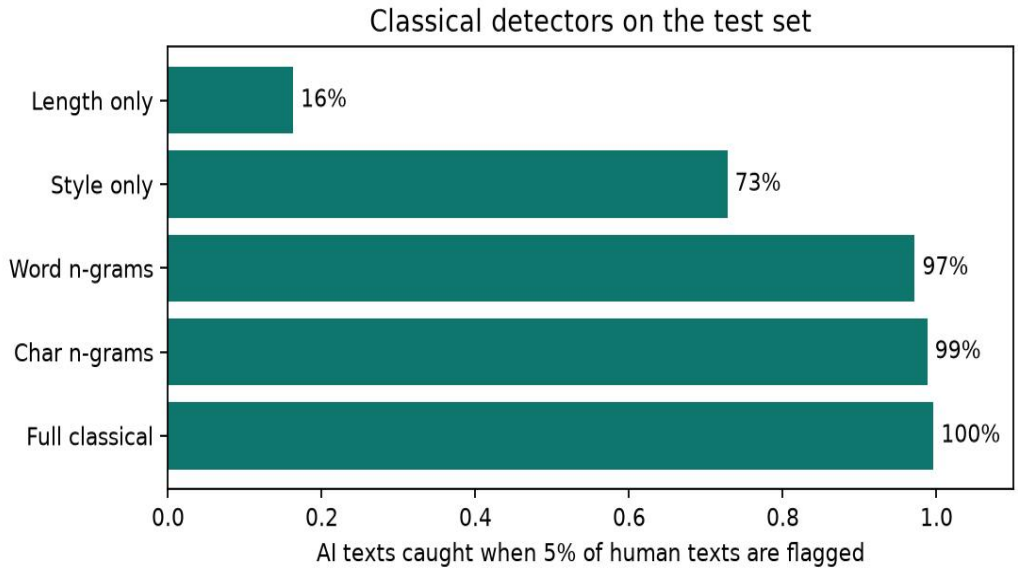


Figure 2. Accuracy and AUROC across feature combinations.

6. Explanations

Beyond accuracy, a goal of this project was to make sure the model's stated reasons are faithful to what it actually used — not just plausible-sounding. A deletion test checks this directly: deleting the five words the explainer ranks highest (by absolute SHAP value) drops the model's predicted confidence by 0.45 on average, versus only 0.15 when five random words are deleted instead. The roughly 3× gap shows the highlighted words are genuinely load-bearing for the prediction, not decorative.

Looking at which stylometric signals matter most globally (Figure 3), human reviews tend to use repeated punctuation ("..", "!!"), informal spelling variants, and emoji; AI-generated reviews tend to use tidier punctuation and somewhat

longer words and sentences. This matches an intuitive expectation — informal, imperfect writing habits are harder for current generators to imitate than clean grammar — and gives a second, independent signal beyond raw lexical overlap.

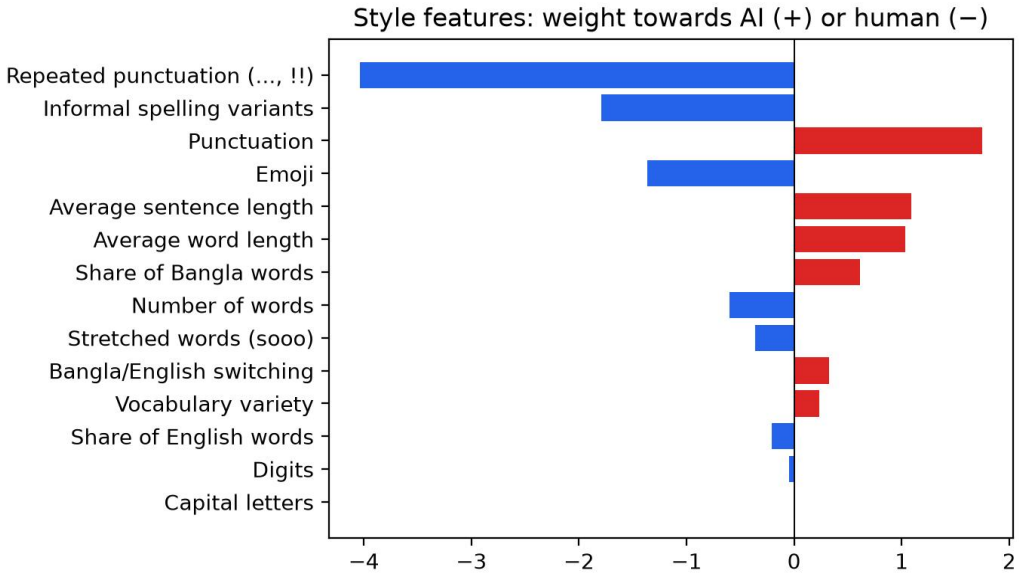


Figure 3. Writing-style signals, ranked by contribution to the AI/human decision.

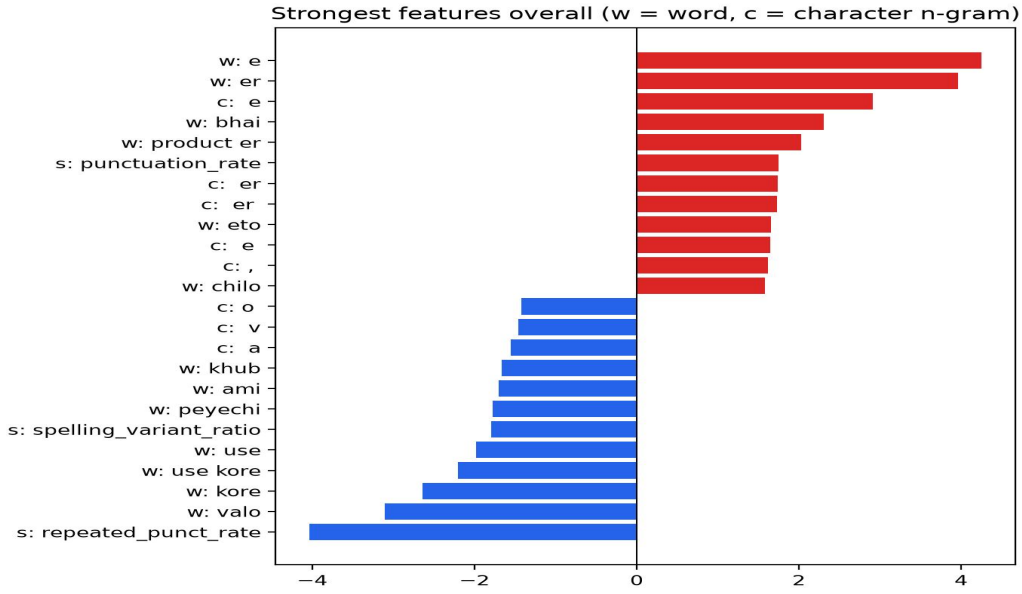


Figure 4. Top word/character features by model coefficient.

7. Robustness

A natural concern for any supervised AI-text detector is generalization to AI models it was not trained on — new generators appear constantly, and a detector trained on a fixed set of them should not be trusted blindly on new ones. A leave-one-generator-out test trains on two of the three AI generators and tests on the held-out third:

Left-out generator	AUROC	AI caught at 5% false flags
Claude	0.938	61.0%
ChatGPT	0.978	91.1%
Gemini	0.996	99.5%

The detector's accuracy drops sharply when Claude is the held-out generator — AUROC falls from 0.996 on the main test set to 0.938, and the AI-catch rate at the 5% false-flag threshold drops from 99.6% to 61.0%. This is the project's clearest limitation: the detector generalizes well to ChatGPT- and Gemini- style writing it has never directly seen, but Claude's Banglish style appears to be distinct enough that unseen-Claude text is substantially harder to catch. In practice this means the detector should be retrained or re-validated whenever it needs to cover a new or updated AI model, and its score should never be treated as proof against a generator it was not validated on.

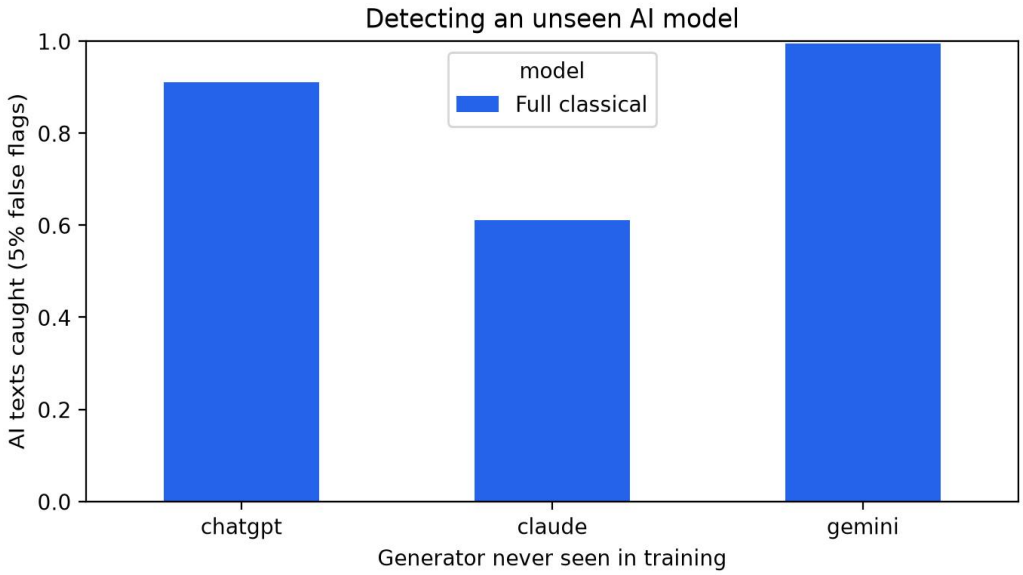


Figure 5. AUROC and AI-catch rate when each generator is held out of training.

8. Limitations and Future Work

- The human-text side of the dataset is entirely product reviews; the model's reliability on other genres of Banglish writing (chat, social media, essays) is untested.
- Texts under roughly 8 words give unreliable predictions, and the deployed app warns the user about this directly.
- Paraphrased or lightly human-edited AI text has not been tested; this is a realistic adversarial case the current evaluation does not cover.
- Only 188 Gemini texts were available, the smallest of the three AI sources — more data per generator, and from more generators, would make both the accuracy numbers and the robustness test more reliable.
- As shown in Section 7, the detector generalizes unevenly across AI generators and should not be assumed to work on a generator it has not been validated against.

If I continued this project, I would first collect balanced data from more generators and from several Banglish genres, then test paraphrased and lightly human-edited AI text. The leave-one-generator-out results show that generator coverage is the most immediate threat to reliability, while the single product-review domain limits what can be claimed about general Banglish writing. A broader evaluation should therefore come before adding a more complex model.

9. Ethics

An AI-text detector carries real risk of harm if misused: a false positive can be used to wrongly accuse a real person of using AI, with consequences ranging from a failed assignment to reputational damage. This project treats that risk as a design constraint rather than a footnote: the deployed app always displays a notice that the score is an estimate, can be wrong, and must never be used on its own to accuse anyone of using AI; it separately flags short texts (under 8 words) as unreliable; and every prediction is shown together with its evidence (highlighted words and style signals) specifically so a human reviewer can sanity-check the model rather than deferring to it.

The same transparency goal motivated the faithfulness test in Section 6 — an explanation that is not demonstrably tied to the model's actual behavior would be worse than no explanation at all, since it could lend false credibility to a wrong decision.

References

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